

Water Quality C - MVSO WQ Div C - Monta Vista Science Olympiad (Div. C) - 11-14-2025
Student Answer Sheet (/monta-vista/SM/AnswerSheet?tid=000GCQ)

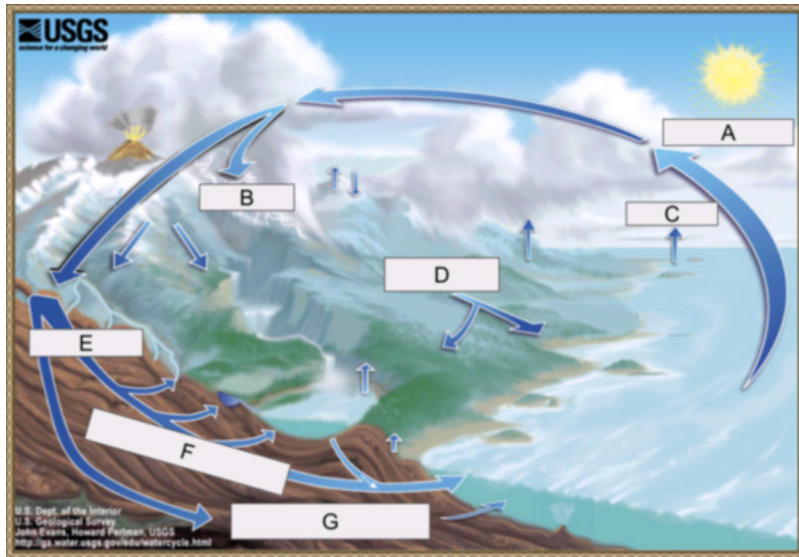
Instructions (shown before students start the test)

- For fill in the blank questions, make sure there are no extra spaces and spelling is correct as they are auto graded.
- For free response questions, no need to write in complete sentences. Bullet points are quicker to write and clearer/easier for me to grade. Just make sure that all key points are mentioned to show understanding.
- For questions involving calculations, the correct answer is sufficient to earn full points. However, showing work may receive partial credit even if the final answer is wrong.
- Good luck!

Introduction (shown after students start the test)

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Use the following image for questions 1 to 3:



1. (3.00 pts) Fill in the following blanks: A B C

2. (3.00 pts) Fill in the following blanks: D E F

3. (1.00 pts) Fill in the following blank: G

4. (3.00 pts) Name and describe 3 unique properties of water.

5. (4.00 pts) Explain how fertilizer can cause eutrophication and its effects.

6. (2.00 pts) Explain lake turnover. How does this impact the water quality of the lake?

7. (1.00 pts) What is Reynolds number?

- ☐ A) A measure of biodiversity for aquatic ecosystems.
- ☐ B) A quantity that helps determine laminar and turbulent flow.
- ☐ C) A measure of specific heat transfer between water and solid materials.
- ☐ D) The ratio of inertial and surface tension forces between water molecules.

8. (1.00 pts) What specific adaptations in the estuary ecosystem allow **plants** to survive?

9. (2.00 pts) What specific adaptations in the estuary ecosystem allow **animals** to survive?

10. (1.50 pts) Which of the following are required by law for all municipal wastewater treatment plants in the US?

(Mark **ALL** correct answers)

- ☐ A) Primary Treatment
- ☐ B) Secondary Treatment

- ☐ C) Tertiary Treatment
- ☐ D) Quaternary Treatment

11. (2.00 pts) Describe the difference between screening, filtration, grit removal, and backwashing.

12. (1.50 pts) Water from which of the following fall under the category of gray water?

(Mark **ALL** correct answers)

- ☐ A) Showers and bathtubs
- ☐ B) Drinking water
- ☐ C) Sinks
- ☐ D) Washing machines
- ☐ E) Toilets

13. (1.00 pts) Which of the following orders is correct?

- ☐ A) Filtration → Flocculation → Disinfection → Coagulation
- ☐ B) Coagulation → Sedimentation → Filtration → Disinfection
- ☐ C) Disinfection → Filtration → Coagulation → Sedimentation
- ☐ D) Aeration → Filtration → Coagulation → Disinfection
- ☐ E) Flocculation → Coagulation → Fluoridation → Disinfection

14. (1.50 pts) What chemical is commonly used for coagulation and flocculation?

(Mark **ALL** correct answers)

- ☐ A) Aluminum sulfate
- ☐ B) Ferric chloride
- ☐ C) Sodium Chloride
- ☐ D) Calcium carbonate
- ☐ E) Ammonia

15. (1.00 pts) Turbidity of drinking water should never exceed _____.

- ☐ A) 0.5 NTU
- ☐ B) 1 NTU
- ☐ C) 3 NTU
- ☐ D) 5 NTU
- ☐ E) 10 NTU
- ☐ F) 50 NTU

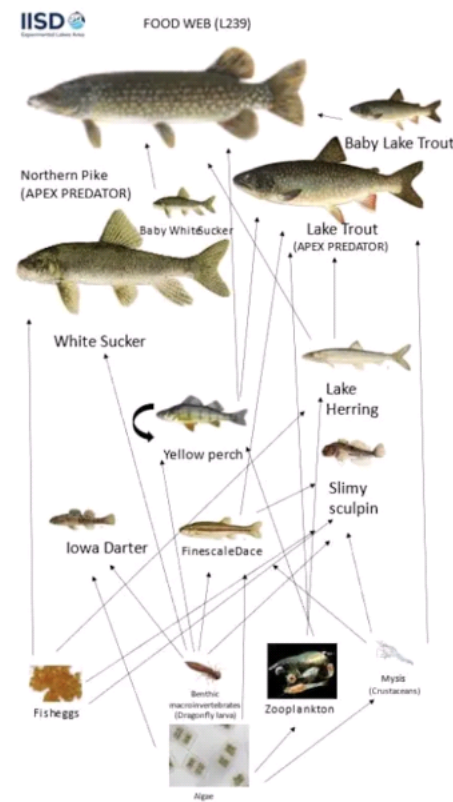
16. (2.00 pts) Which cations enhance flocculation?

(Mark **ALL** correct answers)

- ☐ A) Ca^{2+}
- ☐ B) Cu^{2+}
- ☐ C) Al^{3+}
- ☐ D) Fe^{3+}
- ☐ E) Mg^{2+}
- ☐ F) Na^+

17. (1.00 pts) What is activated sludge?

- ☐ A) Sludge that has been chlorinated/removed.
- ☐ B) Sludge containing active microorganisms used to treat wastewater.
- ☐ C) Sludge decomposed by bacteria in an anaerobic digester.
- ☐ D) Sludge residue that has become thicker as a result of an industrial refining process.



18. (1.00 pts) A species can only occupy one trophic level at a time.

- ☐ True
- ☐ False

19. (1.00 pts)

Assume algae → mysis → finescale dace → lake trout. If the algae starts with 100 kJ of energy, how much energy does the lake trout receive? Answer in joules and do not include units.

20. (1.00 pts) Explain the difference between bioaccumulation and biomagnification.**21. (4.00 pts)**

Aquatic ecosystems such as rivers, lakes, and estuaries are highly dynamic and influenced by both natural processes and human activities. Factors such as nutrient runoff, invasive species, and thermal pollution can alter water chemistry and biodiversity, leading to changes in ecosystem stability and productivity. Explain how traditional ecological knowledge (TEK) can be used to analyze anthropogenic impacts.

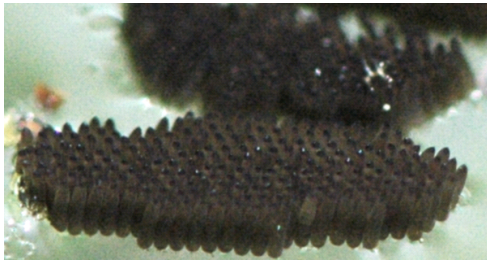
Identify the following organisms.

22. (1.00 pts)

23. (1.00 pts)



24. (1.00 pts)



25. (1.00 pts)



26. (1.00 pts)



27. (1.00 pts)



28. (1.00 pts)



29. (1.00 pts)



30. (1.00 pts)



31. (1.00 pts) Organisms need to eat to survive. However, many insect species stop eating after becoming an adult. Why is this?

32. (3.00 pts) What is the difference between complete and incomplete metamorphosis?

33. (1.00 pts) Flatworms reproduce through _____.

- ☐ A) Sexual reproduction
- ☐ B) Asexual reproduction
- ☐ C) Neither
- ☐ D) Both

34. (1.00 pts) Just like earthworms, leeches are annelids, meaning that they _____.

- ☐ A) have bilateral symmetry
- ☐ B) lack a spine and act as decomposers
- ☐ C) are parasites and suck blood from their host
- ☐ D) have segmented bodies that allows for flexible movement

35. (2.00 pts) Why do blood midges have their distinct red color?

36. (1.00 pts) Which property of water allows water striders to float on water?

- ☐ A) High density of water
- ☐ B) Surface tension of water
- ☐ C) Capillary action
- ☐ D) Cohesion of water molecules

37. (1.00 pts) How does the water boatman breath air underwater?

- ☐ A) They receive oxygen through their skin in pockets at the tip of the abdomen.
- ☐ B) They carry air bubbles on their body's surface or under their wings while they are underwater.
- ☐ C) They don't. Adults must rise to the surface to receive oxygen from the air.
- ☐ D) Air filters on the side of its body allows them to receive oxygen.

38. (1.50 pts) Which of the following are part of the Hemiptera order of insects?

(Mark **ALL** correct answers)

- ☐ A) Blackfly
- ☐ B) Back swimmer
- ☐ C) Dragonfly
- ☐ D) Mayfly

39. (1.00 pts) The male adults of which species have long, curving mandibles, easily distinguishing them from other insects?

- ☐ A) Water scorpion
- ☐ B) Predacious diving beetle
- ☐ C) Dobsonfly
- ☐ D) Mayfly

40. (3.00 pts) Name two examples of ways sediments can pollute waterways.

41. (1.50 pts) Positive impacts of moderate sedimentation include:

(Mark **ALL** correct answers)

- ☐ A) Nutrient input

- ☐ B) Improved water quality
- ☐ C) Surface runoff
- ☐ D) Natural lake succession

42. (1.50 pts) What is the normal pH range of rivers.

(Mark **ALL** correct answers)

- ☐ A) 5.5-6.5
- ☐ B) 6.5-7.5
- ☐ C) 7.5-8.5
- ☐ D) 8.5-9.5

43. (1.50 pts) What substance is commonly dissolved to increase the alkalinity of a lake?

44. (1.00 pts)

John measures the turbidity of a water sample by measuring the incident light scattered at right angles from the sample. What method of measuring turbidity is John using?

- ☐ A) ISO 7027-1
- ☐ B) Jackson Candle Method
- ☐ C) Secchi Disk
- ☐ D) Not a real method.

45. (1.00 pts) Which of the following is NOT a cause or effect of supersaturation:

- ☐ A) Photosynthesis
- ☐ B) Hypoxia
- ☐ C) Block blood flow
- ☐ D) Decompression sickness

46. (1.00 pts) Which of the following is an example of thermal enrichment:

- ☐ A) Trout that prefer warmer environments.
- ☐ B) Manatees using warm water near power plants.
- ☐ C) Bass respond positively to increased nutrients.
- ☐ D) Duckweed which spike in population growth.

47. (1.50 pts) Which diseases are linked with fecal coliform:

(Mark **ALL** correct answers)

- ☐ A) Cholera
- ☐ B) Salmonella

- ☐ C) Dysentery
- ☐ D) Hepatitis
- ☐ E) Typhoid

48. (1.50 pts) Select two ways total dissolved solids can be measured.

(Mark **ALL** correct answers)

- ☐ A) Titration
- ☐ B) Gravimetric
- ☐ C) Temperature
- ☐ D) Conductivity

49. (1.00 pts) Total suspended solids are not considered a conventional pollutant, but total dissolved solids are.

- ☐ True ☐ False

50. (5.00 pts)

You are analyzing a river water sample collected below a wastewater outfall. The BOD test setup includes two 300 mL BOD bottles filled with dilution water and two 300 mL BOD bottles containing 6 mL of river water sample + dilution water.

Do measurements were taken initially (Day 0) and after 5 days (Day 5)

	Initial DO (mg/L)	Final DO (mg/L)
Blank 1	8.15	8.05
Blank 2	8.20	8.10
Sample 1	8.10	3.25
Sample 2	8.05	3.15

Calculate the BOD₅ of the river water sample (mg/L)

51. (3.00 pts) After the temperature increases, state whether the following will increase, decrease, or have no effect.

1. Dissolved oxygen 2. Total solids 3. Phosphates

52. (1.00 pts) The Winkler Method uses what chemical practice to determine what quality of water?

- ☐ A) Redox reaction, biological oxygen demand
- ☐ B) Redox reaction, dissolved oxygen
- ☐ C) Titration, biological oxygen demand

- ☐ D) Titration, dissolved oxygen

53. (1.00 pts) Which of the following is not a source of phosphorus?

- ☐ A) Pesticides
☐ B) Cleaning compounds
☐ C) Wastewater from sewage treatment
☐ D) Erosion
☐ E) Mining
☐ F) Gas exchange

54. (1.00 pts) Which form of phosphorus is readily available for biological processes without further breakdown in aquatic organisms?

- ☐ A) Phosphates
☐ B) Orthophosphates
☐ C) Polyphosphates
☐ D) Biphosphorus

55. (4.50 pts) What is the equation to determine pH given the concentration of H^+ ? Calculate the pH of a solution with a H^+ concentration of 4×10^{-8} M.

Congratulations on finishing MVSO Invitational's water quality test! Good luck on all your other events!

If you have any questions or would like to provide feedback:

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